

3-Mark Questions & Answers MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Draw and explain the functional pin diagram of 8086 (any 6-8 important pins).

1. **AD0 - AD15 (Pins 2-16, 39):** Multiplexed Address/Data lines carrying 16-bit data or lower 16-bit memory address.

2. **A16 - A19 (Pins 17-20):** 4-bit bus extension for memory addresses up to 1 MB.

Q2. Explain memory segmentation in 8086 with a diagram.

1 MB Memory Segmentation Diagram

Q3. Explain the concept of banking (memory banking) in 8086.

1 MB Memory Banking Diagram

Q4. Differentiate between minimum mode and maximum mode operation of 8086.

| Feature | Minimum Mode | Maximum Mode |

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Q5. Explain the need for demultiplexing the address/data bus in 8086.

1. Pins **AD0 - AD15** carry memory addresses during the first clock cycle (**T₁**) and data during **T₂, T₃, T₄**.

2. With